資料視覺化
Data Visualization

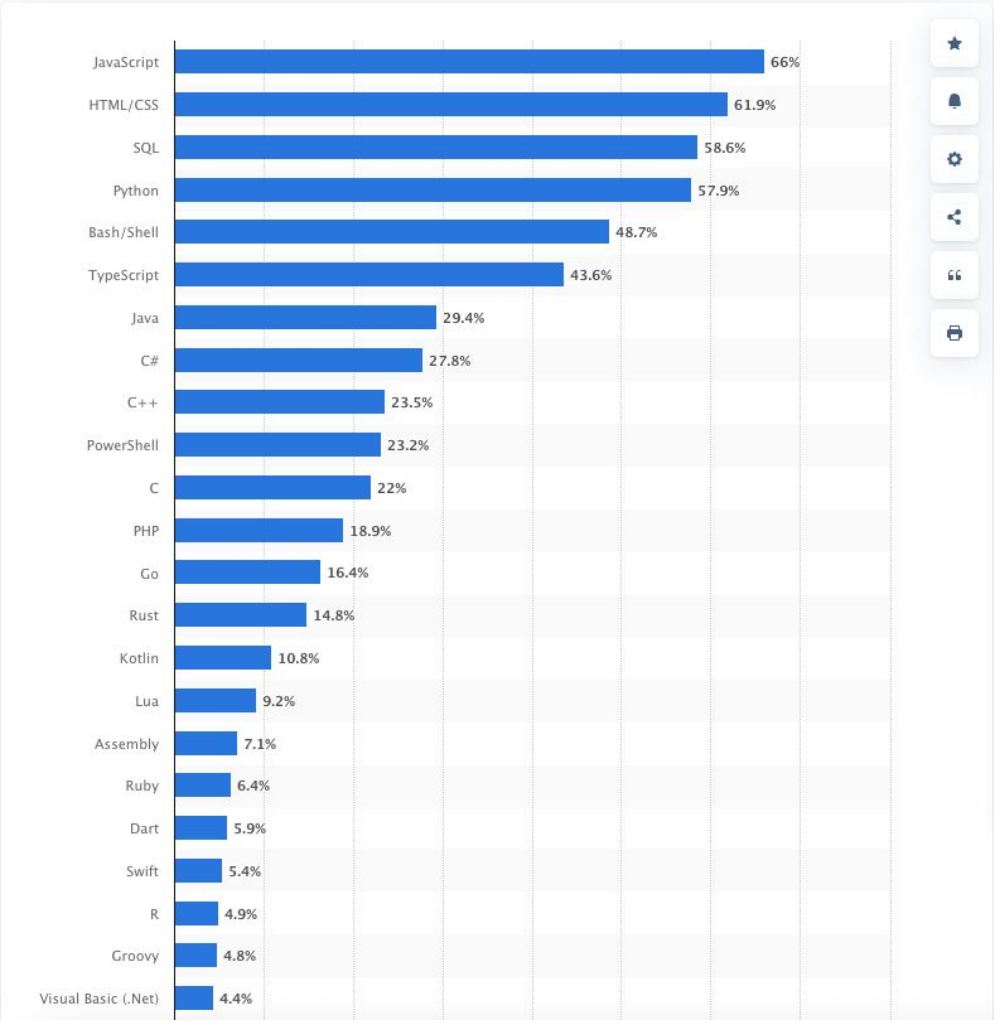
Week 2: Getting Started with Python

The Structure of our Classes

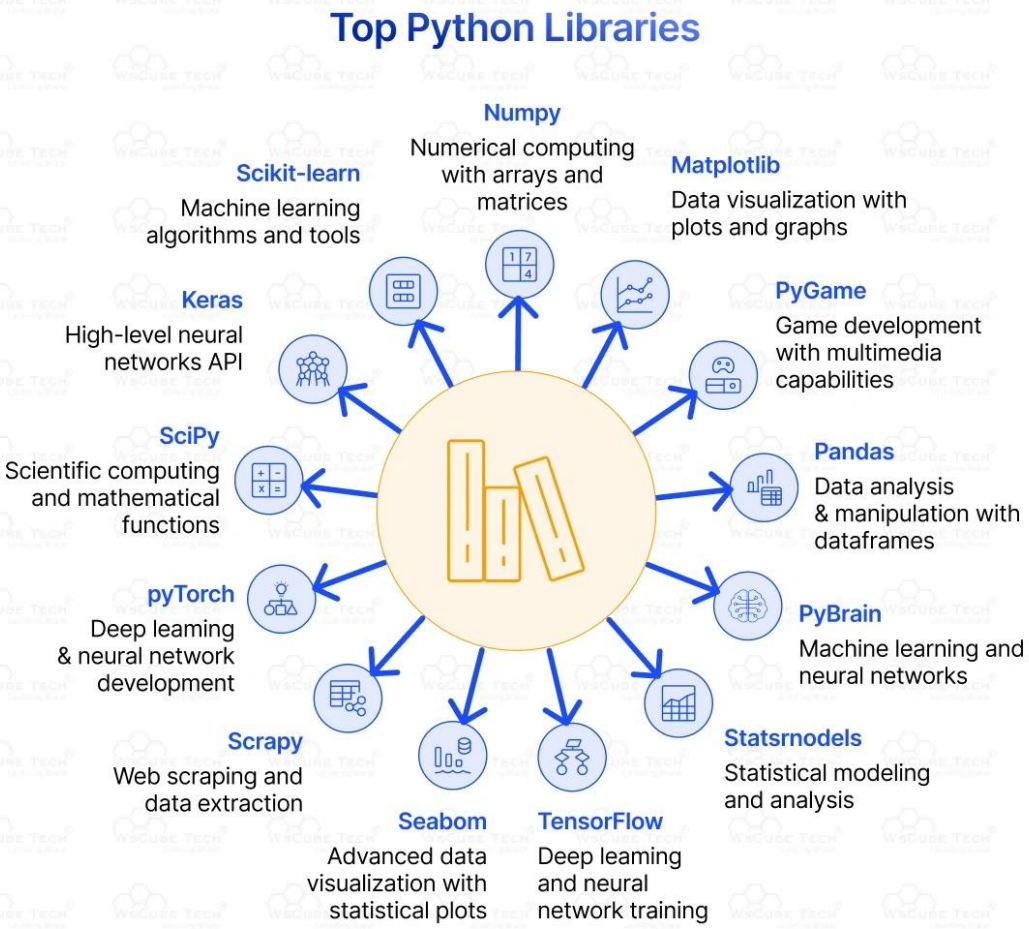
- Lecture (me talking)
- Practical (me coding)
- Challenges (you coding)

Why Python?

Python is one of the most used programming languages among developers worldwide



Strong Ecosystem of Libraries



WsCube Tech®
Upskilling Bharat

Extremely Scalable

Streamlit

Flask
web development,
one drop at a time

django

Reproducibility

- Python lets us automate repeated tasks
- Once a script is written, we can rerun it whenever new data arrives
- Saves time and ensures consistent results

Open Science

- “Open science is a global movement that aims to make scientific research and its outcomes freely accessible to everyone”
- Transparency, sharing, and inclusivity
- Python is open source
- Python makes it easier for:
  - Open data
  - Open code
  - Publishing results in ways others can read, check, and reproduce

Strong Community and Support

- Python has a large global community
- Thousands, if not millions, free tutorials, examples, and open-source
- Check this out: https://www.py4e.com/

Some practical things...

Relationship between Python and Jupyter

If you are using something else...

- https://code.visualstudio.com/docs/datascience/jupyter-notebooks
- https://www.anaconda.com/

You can even run it online!

- https://colab.research.google.com/

If you are hardcode enough…
(don’t try this at home)

home.py

```
def check_password():
    """Returns `True` if the user had the correct password."""

    def password_entered():
        """Checks whether a password entered by the user is correct."""
        # Use environment variable for the password
        if hmac.compare_digest(st.session_state["password"], APP_PASSWORD):
            st.session_state["password_correct"] = True
            del st.session_state["password"]  # Don't store the password.
        else:
            st.session_state["password_correct"] = False

    # Return True if the password is validated.
    if st.session_state.get("password_correct", False):
        return True

    col1, col2, col3 = st.columns([5, 1, 5], vertical_alignment="top")

    # Show input for password.
    col1.text_input(
        label = "", type="password", on_change=password_entered, key="password",
placeholder="Enter password", label_visibility="collapsed"
    )
    with col2:
        st.markdown("""
        <p style='text-align: center;'>
            <i>or</i>
        </p>
        """, unsafe_allow_html=True)
    col3.link_button("Request password", url="https://forms.office.com/e/3YxcZeoNrK"
```

When I code

- It is highly recommended to code along with me
- You could let me know when you come across and error right away
- Trust me, you don’t know what you don’t know until you start typing…

If you come across an error...

- Error is the best learning opportunity
- The more errors you make, the faster you learn (trust me on this)
- I want you to share me the codes so that I can reproduce the error and solve it together
- https://pad.carpentries.org/nycudata

Postit System

- Pink →

  I NEED HELP!!!!

- Green →

  I am alright.

Don’t be like this...

Time for Python

Data Types

Numbers

- Int, or integer, is a whole number, positive or negative, without decimals, of unlimited length.
- Float, or "floating point number" is a number, positive or negative, containing one or more decimals.
- You can convert from one type to another with int() and float()

String

- Strings in python are surrounded by either single quotation marks, or double quotation marks
- 'hello' is the same as "hello"
- You can also do this: "It's alright"
- You can assign a multiline string to a variable by using three quotes:

```
texts = """Lorem ipsum dolor sit amet,
           consectetur adipiscing elit,
           sed do eiusmod tempor incididunt
           ut labore et dolore magna aliqua."""
```

String Methods

Check this:

https://www.w3schools.com/python/python_strings_methods.asp

List

```
mylist = ["apple", "banana", "cherry"]
```

- Store multiple items in a single object
- Ordered, changeable, and allow duplicate values

List Methods

Method	Description
append()	Adds an element at the end of the list
clear()	Removes all the elements from the list
copy()	Returns a copy of the list
count()	Returns the number of elements with the specified value
extend()	Add the elements of a list (or any iterable), to the end of the current list
index()	Returns the index of the first element with the specified value
insert()	Adds an element at the specified position
pop()	Removes the element at the specified position
remove()	Removes the item with the specified value
reverse()	Reverses the order of the list
sort()	Sorts the list

Dictionary

```
thisdict = {
  "brand": "Ford",
  "model": "Mustang",
  "year": 1964
}
```

- Store data values in key:value pairs.
- Ordered, changeable and *DO NOT* allow duplicates

Dictionary Methods

Method	Description
clear()	Removes all the elements from the dictionary
copy()	Returns a copy of the dictionary
fromkeys()	Returns a dictionary with the specified keys and value
get()	Returns the value of the specified key
items()	Returns a list containing a tuple for each key value pair
keys()	Returns a list containing the dictionary's keys
pop()	Removes the element with the specified key
popitem()	Removes the last inserted key-value pair
setdefault()	Returns the value of the specified key. If the key does not exist: insert the key, with the specified value
update()	Updates the dictionary with the specified key-value pairs
values()	Returns a list of all the values in the dictionary